



VEKTOR RESEARCH SERIES · 2026

Multi-Currency Portfolio Infrastructure

*How Vektor handles daily FX rate collection, currency-adjusted portfolio valuation,
and cross-border mandate management — for any currency pair.*

For professional and institutional investors

IP-WP-FX-260501-1.0

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ABSTRACT

Multi-currency is treated as a configuration option in most portfolio platforms. In Vektor, it is a first-class design requirement, built in from the first prototype. Wealth teams managing clients across multiple currencies face a persistent infrastructure problem: portfolio valuations, performance calculations, and allocation decisions that span currency boundaries require daily FX rate data, consistent currency conversion logic, and a portfolio architecture that treats multi-currency mandates as a first-class use case rather than an afterthought. This paper describes Vektor's multi-currency infrastructure: daily FX collection, currency-adjusted valuation, and the platform's support for mandates spanning any combination of listed equity markets and base currencies.

KEY TAKEAWAYS

- Daily FX rate collection at 8 AM SGT ensures all portfolio valuations use current exchange rates — not rates from the previous week or quarter.
- Currency conversion is applied at the position level — enabling accurate per-instrument P&L; attribution across currency boundaries.
- Any base currency can be configured per client — the same strategy can be run for an SGD client and a USD client simultaneously.
- FX rate history supports backwards-looking performance analysis in any reporting currency.

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1. THE MULTI-CURRENCY PROBLEM IN WEALTH MANAGEMENT

Why currency handling is not a peripheral feature — it is a core infrastructure requirement.

A boutique family office managing clients across Singapore, the UK, and the US faces a currency challenge on every portfolio decision. An allocation to Singapore equities for a USD-base client requires real-time SGD/USD conversion for accurate position valuation. A performance report for a GBP-base client holding a mixed SGX/LSE portfolio requires daily FX rates applied at the position level for each trading day in the reporting period. A rebalancing decision requires knowing the current FX-adjusted weight of each position — not the weight at the time of original allocation, which may have drifted materially with currency movements.

None of these requirements are exotic. They are the daily operational reality of multi-currency wealth management.

The alternative — converting currencies manually using bank rates or periodic batch processes — introduces latency, inconsistency, and the risk of valuation errors that compound over time. A platform that treats multi-currency as a first-class feature eliminates this risk entirely.

2. DAILY FX RATE COLLECTION

How Vektor collects and stores exchange rates — automatically, every market day.

Vektor collects daily foreign exchange rates from Yahoo Finance at 8 AM SGT every market day — the same time as the daily OHLCV price data refresh. Rates are stored with full history, enabling backwards-looking FX-adjusted performance calculations across any time period. The collection covers all FX pairs required by configured mandates. New currency pairs are added to the collection schedule automatically when a new mandate with a different base currency is configured.

PLATFORM INTERFACE

VEKTOR

PM Portfolio Manager
demo@investpuppy.com Logout

Dashboard
Overview & Analytics

CLIENT MANAGEMENT

Clients
Profiles & KYC

Onboarding
New Client Setup

PORTFOLIO MANAGEMENT

Portfolios
Portfolio Overview

Strategies
Strategy Composition

Allocations
Asset Allocation

Rebalancing
Rebalance Portfolios

TRADING & ORDERS

Orders
Trade Orders

Executions
Trade Executions

Settlements
Trade Settlements

MARKET DATA

Instruments
Securities Master

Exchange Rates
FX Rates (SGD/USD)

Instruments

Search by symbol, name, or sector...

Showing 10 of 10 instruments

Symbol	Name	Type	Exchange	Sector	Market Cap	Actions
595	GKE Corporation Limited SGD	equity	SGX	Industrials	\$70M	Prices
856	Yangzijiang Shipbuilding (Holdings) Ltd. SGD	equity	SGX	Industrials	\$15.7B	Prices
8U0U	Frasers Logistics & Commercial Trust SGD	equity	SGX	Real Estate	\$3.8B	Prices
C67	Jardine Cycle & Carriage Limited SGD	equity	SGX	Industrials	\$13.0B	Prices
C09	City Developments Limited SGD	equity	SGX	Real Estate	\$7.4B	Prices
C2PU	Parkway Life Real Estate Investment Trust SGD	equity	SGX	Real Estate	\$2.6B	Prices
D85	DBS Group Holdings Ltd SGD	equity	SGX	Financial Services	\$166.5B	Prices
039	Oversea-Chinese Banking Corporation Limited SGD	equity	SGX	Financial Services	\$98.4B	Prices
563	Singapore Technologies Engineering Ltd SGD	equity	SGX	Industrials	\$33.3B	Prices
U10	UOB-Kay Hian Holdings Limited SGD	equity	SGX	Financial Services	\$4.1B	Prices

Instruments master — securities listed with exchange, currency, sector, and market cap. Multi-currency mandates configure the base currency per client; the platform handles conversion automatically.

3. CURRENCY-ADJUSTED PORTFOLIO VALUATION

How FX rates are applied at the position level to produce accurate cross-currency valuations.

Portfolio valuation in the client's base currency requires applying the relevant FX rate to each position denominated in a foreign currency. Vektor applies currency conversion at the position level — not at the portfolio aggregate level — ensuring that each holding's contribution to portfolio value is accurately stated in the reporting currency.

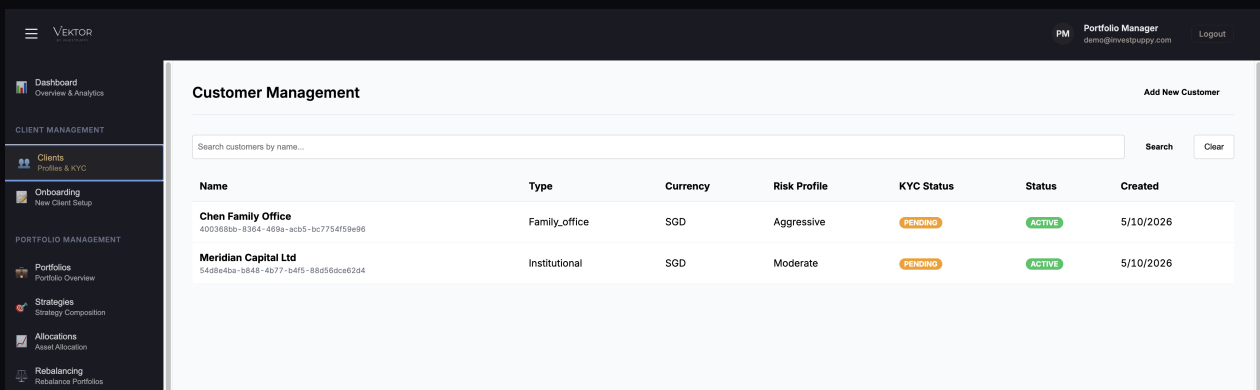
This distinction matters for performance attribution: a position that gained 5% in local currency terms but lost 3% due to currency movement contributed only 2% to portfolio return in the base currency. Aggregate-level conversion masks this — position-level conversion reveals it.

4. PER-CLIENT BASE CURRENCY CONFIGURATION

How the same strategy serves clients with different base currencies simultaneously.

Each client in Vektor has a configured base currency. The same investment strategy — for example, the SGP Equity Strategy holding SGX-listed securities in SGD — can simultaneously serve an SGD client and a USD client. For the SGD client, all valuations are in SGD. For the USD client, the same holdings are valued in USD using the daily SGD/USD rate. Capital allocation for the USD client is calculated in USD and converted to SGD at the time of position sizing. No manual conversion is required at any stage.

PLATFORM INTERFACE



The screenshot shows the Vektor Platform Interface. The top navigation bar includes the Vektor logo, a PM (Portfolio Manager) role indicator, the user email (demo@investpuppy.com), and a Logout button. The left sidebar contains a menu with options: Dashboard (Overview & Analytics), CLIENT MANAGEMENT (Clients Profiles & KYC, Onboarding New Client Setup), PORTFOLIO MANAGEMENT (Portfolios Portfolio Overview, Strategies Strategy Composition, Allocations Asset Allocation, Rebalancing Rebalance Portfolios), and a Rebalancing button. The main content area is titled 'Customer Management' and includes an 'Add New Customer' link. Below this is a search bar with the placeholder text 'Search customers by name...'. A table lists two clients:

Name	Type	Currency	Risk Profile	KYC Status	Status	Created
Chen Family Office 400388bb-8364-498a-acb5-bc7754f59e96	Family_office	SGD	Aggressive	PENDING	ACTIVE	5/10/2026
Meridian Capital Ltd 54ab4fe4-b04d-4b77-b4f5-88d5dc622d4	Institutional	SGD	Moderate	PENDING	ACTIVE	5/10/2026

Customer management — clients configured with individual base currencies. The same strategy can be assigned to multiple clients with different currencies; valuation and reporting are handled per client automatically.

5. GENERALISING BEYOND SGD/USD

How the multi-currency infrastructure scales to any market and any currency pair.

The multi-currency infrastructure described in this paper is not specific to SGD/USD. Any listed equity market can be configured — London-listed industrials in GBP, US equities in USD, European equities in EUR — and any base currency can be set per client. The FX collection schedule expands automatically as new currency pairs are required. The position-level conversion logic applies identically regardless of the currency pair. The platform is currency-agnostic by design, consistent with its market-agnostic architecture.

6. CONCLUSION

Multi-currency support is not a feature to be added later. It is a requirement that must be designed in from the start.

Vektor's daily FX collection, position-level currency conversion, and per-client base currency configuration provide the infrastructure that wealth teams managing cross-currency mandates require — without the manual processes, batch conversions, and valuation inconsistencies that characterise currency handling in spreadsheet-based workflows.

Any currency pair. Any exchange. The configuration changes. The rigour does not.

This paper describes Steps 10–11 of the Vektor workflow. For the full platform overview, see the Platform Brochure. For portfolio construction methodology, see WP-01. All documents at investpuppy.com.